

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology (B.Tech) (EE)

University Examination November 2010

EE-208 Micro Electronics & Integrated Circuit

Date: 30/11/10, Tuesday

Time: 10:30 a.m. to 1.30p.m.

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Figure to the right indicates full marks.
3. Assume suitable data, if necessary and mention it.
4. Attempt both section separately.

SECTION-I

- Q.1 (a) List out the ideal characteristics of op-amp with explanation. 04
- (b) Draw and Explain the block diagram of op-amp? 06
- (c) Define the following terms. 05
- (1) Input offset voltage (2) Input bias current (3) CMRR
- (4) Slew Rate (5) Bandwidth
- Q.2 (a) Explain integrator and differentiator using op-amp. 04
- (b) Draw the circuit of a voltage follower using op-amp and prove that its gain is exactly equal to unity. 03
- (c) Design first order active low pass filter with cut-off frequency 1kHz and pass band gain of 2. 03
- OR
- Q.2 (a) Explain first order low pass butter worth filter with necessary circuit diagram. 04
- (b) Explain saw tooth wave generator using op-amp. 03
- (c) Explain types of filters in detail. 03
- Q.3 (a) Explain adjustable voltage regulator and write the advantages of adjustable regulator over fixed voltage regulator. 04
- (b) Explain astable multivibrator using IC-555 with suitable waveform and derive the equation of total time T. 06
- OR
- Q.3 (a) Design an adjustable voltage regulator to satisfy the following specifications. 04
- Output voltage: $V_o = 5$ to $12V$
- Output current: $I_o = 1.0A$
- Voltage regulator is LM317
- For LM317: $I_{adj} = 100 \mu A$ (maximum)
- Assume $R_1 = 240 \Omega$

FOR REFERENCE

Q.3 (b) Write short note:

06

- I) IC-555 as a timer.
- II) Fixed voltage regulator

SECTION – II

Q.4 (a) Draw and explain about common-base connection for NPN transistor with its input-output characteristics. 08

(b) Explain Voltage divider biasing method. How it provides stability to the transistor. 07

Q.5 (a) Write difference between JFET and bipolar transistor. 04

(b) Define Pinch-off voltage in JFET. 03

(c) Explain working principle of JFET. 03

OR

Q.5 (a) Explain self-bias method in JFET. 04

(b) Define shorted-gate drain current in JFET. 03

(c) Explain working principle of MOSFET. 03

Q.6 (a) A transistor is connected in common emitter(CE) configuration in which collector supply is 8V and voltage drop across resistance R_c connected in the collector circuit is 0.5V. The value of $R_c = 800\Omega$. If $\alpha = 0.96$, Determine: (i) collector-emitter voltage (ii) base current. 04

(b) Define the following terms related to transistor 06

- 1) Cut-off point
- 2) Saturation point
- 3). Active region

OR

Q.6 (a) The following quantities are measure in a CE amplifier circuit. 04

(i) With output a.c. short circuited(i.e. $V_{ce}=0$)

$$I_b = 10\mu A; I_c = 1mA, V_{be} = 10mV$$

(ii) With input a.c. open-circuited(i.e. $I_b = 0$)

$$V_{be} = 0.65 mV; I_c = 60 \mu A; V_{ce} = 1V.$$

Determine all the four h-parameters.

(b) Describe h-parameter of a transistor and its circuit performance. 06